

AhamX Bodhi

Where I Becomes Infinite

Residual Connections

Module 3.4: Language Model Head and Training Targets

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Session Overview

What You Will Learn

- The vanishing gradient problem that motivated residual connections
- The residual (skip connection) formula and its gradient properties
- Pre-LN vs Post-LN residual placement
- How residuals enable 96+ layer LLMs
- Residual connections in GPT, Llama, and production models

Concept at a Glance

- **Duration:** 9 minutes
- **Bloom's Level:** B3 (Apply)
- **Difficulty:** L2 (Intermediate)
- **Key Formula:** $\mathbf{x}_{l+1} = \mathbf{x}_l + F(\mathbf{x}_l)$
- **Origin:** ResNet (He et al., 2016)

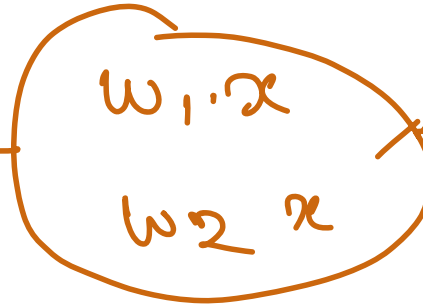
Learning Objectives



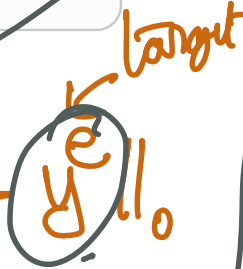
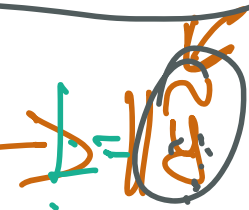
By the End of This Concept You Will Be Able To

- **Explain** why deep networks without residuals suffer from vanishing gradients
- **Write** the residual connection formula and derive how it preserves gradient flow
- **Distinguish** Pre-LN from Post-LN residual placement and state their trade-offs
- **Identify** where residual connections appear in the GPT and Llama block structure
- **Connect** residual connections to training stability and depth scalability

$f_e \rightarrow \text{target}$
 x, y_e



$2x$



min

$\rightarrow 0$

$\Rightarrow e$ will be
min @ $\frac{\partial e}{\partial w_1} = 0$

$\nabla L \Rightarrow 0 \Rightarrow \nabla f$
 $\frac{\partial L}{\partial w_1}$ & $\frac{\partial L}{\partial w_2}$

$$w_i^{k+1} = w_i^k + \eta \left| \frac{\partial L}{\partial w_i} \right|$$

Learning rate
 $\sim 10^{-3}$
 10^{-5}

~~x^{exact}~~ , $\tilde{x}$ - app. of x^{exact}

$$Ax^e = b$$

$$e = x^e - \tilde{x}$$

What is residue? $r = b - A\tilde{x}$

$$Ae = A(x^e - \tilde{x}) = \underline{\underline{Ax^e}} - A\tilde{x} = \underline{\underline{b}} - A\tilde{x}$$

$$Ae = r.$$



The Vanishing Gradient Problem

Why Deep Networks Fail Without Skip Connections

- In a deep network without residuals, the gradient of the loss flows through every layer by the chain rule
- Each layer multiplies the gradient by its local Jacobian $\partial F_l / \partial \mathbf{x}_l$
- If these Jacobians have spectral norm < 1 , the gradient **shrinks exponentially** with depth
- With 50+ layers, gradients arriving at the first few layers are effectively zero
- Result: early layers receive no useful training signal; the model cannot learn

This problem was the primary reason early deep networks (pre-2015) were limited to ~ 10 layers. The transformer architecture uses 96–120+ layers in GPT-4 scale models, which is only possible due to residual connections.



The Solution: Residual Learning (He et al., 2016)

The Key Insight from ResNet

- Instead of learning a mapping $H(\mathbf{x})$ directly, learn the **residual** $F(\mathbf{x}) = H(\mathbf{x}) - \mathbf{x}$
- The layer output becomes: $\mathbf{x}_{l+1} = \mathbf{x}_l + F_l(\mathbf{x}_l)$
- This adds a **skip connection** (also called shortcut connection) bypassing the sub-layer F_l
- If the sub-layer learns $F_l(\mathbf{x}_l) \approx 0$, the output is unchanged — a **safe identity fallback**
- ResNet-152 (2015) demonstrated 8.1% ImageNet top-5 error — 12 points better than previous methods

He et al.'s insight was that it is easier to learn a small correction to the identity than to learn a full mapping from scratch. Residual connections make the **identity function the default**, with the network learning only deviations from it.





Why Residuals Fix Vanishing Gradients

The Gradient Identity Property

- For $\mathbf{x}_{l+1} = \mathbf{x}_l + F_l(\mathbf{x}_l)$, the gradient of loss $\mathcal{L}$ with respect to $\mathbf{x}_l$ is:

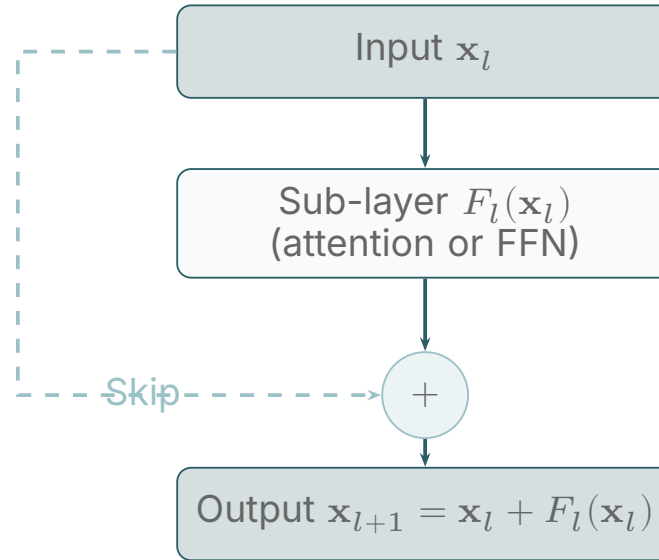
$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}_{l+1}} \left(1 + \frac{\partial F_l}{\partial \mathbf{x}_l} \right)$$

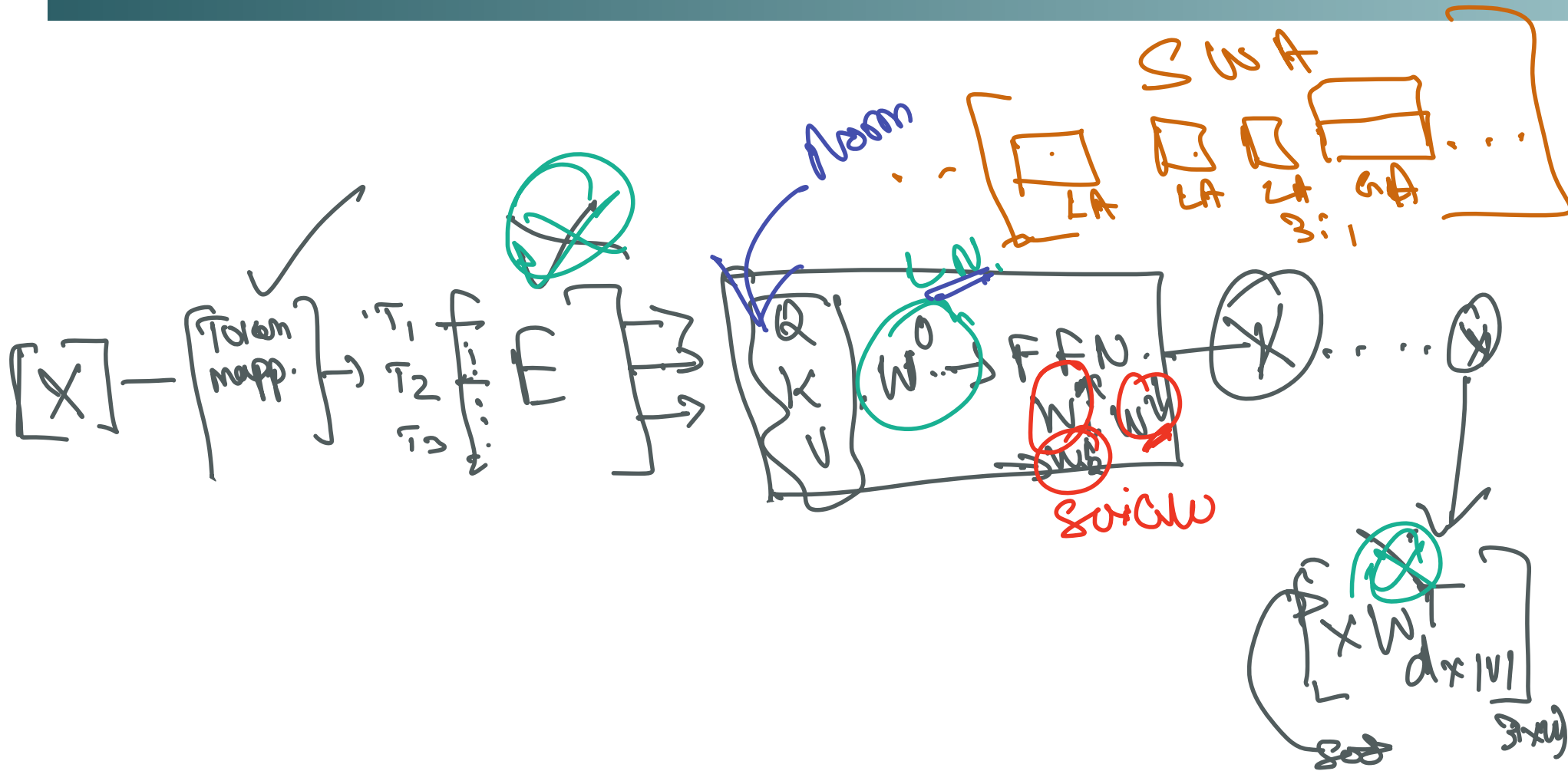
- The **constant 1** in the parentheses means: even if $\partial F_l / \partial \mathbf{x}_l \approx 0$, the gradient is still $\partial \mathcal{L} / \partial \mathbf{x}_{l+1}$
- This **identity path** propagates the full gradient signal regardless of sub-layer depth
- For a chain of L residual layers: $\partial \mathcal{L} / \partial \mathbf{x}_0 = \prod_l (1 + \partial F_l / \partial \mathbf{x}_l)$, which grows if corrections are non-trivial





Residual Connection: Architectural Diagram



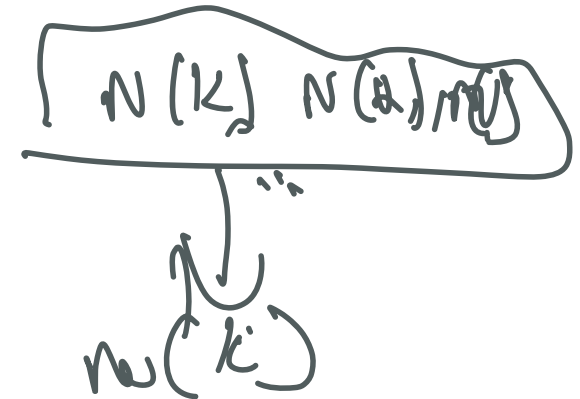




Residuals in the Transformer Block

Two Residuals Per Block

- Every Transformer block contains exactly **two** residual connections
- **After attention sub-layer:** $x' = x + \text{MultiHeadAttn}(\text{Norm}(x))$
- **After FFN sub-layer:** $x'' = x' + \text{FFN}(\text{Norm}(x'))$
- With N transformer blocks, there are $2N$ residual additions in the forward pass
- GPT-2 Large ($N = 36$): 72 residual additions. Estimated GPT-4 ($N \approx 96-120$): 192-240 additions





Pre-LN vs Post-LN: Where to Normalise

Post-LN (Original Transformer)

- $\mathbf{x}' = \text{LayerNorm}(\mathbf{x} + F(\mathbf{x}))$
- Normalisation **after** the residual addition
- Training can be unstable in deep models
- Requires careful learning rate warmup
- Used in: original Transformer (2017), BERT

Pre-LN (Modern Standard)

- $\mathbf{x}' = \mathbf{x} + F(\text{LayerNorm}(\mathbf{x}))$
- Normalisation **before** the sub-layer, inside the residual branch
- More stable training; less sensitive to learning rate
- Gradient flows cleanly through the skip path
- Used in: GPT-2, GPT-3, Llama, Mistral, Gemma

Why Pre-LN Became the Modern Standard

$\rightarrow A, K, V$ -norm



The Mathematical Reason

- In Post-LN, the gradient of the loss *at early layers* passes through the LayerNorm operation
- LayerNorm can amplify or suppress gradients depending on activation magnitudes
- In Pre-LN, the skip connection carries gradients *without any normalisation*, preserving the identity gradient property
- At the end of the model, an additional Final LayerNorm is added before the LM head to stabilise activations

Xiong et al. (2020) formally proved that Pre-LN Transformers have bounded gradients at initialisation, enabling training without warmup. This is why all major LLMs since GPT-2 use Pre-LN.



RMSNorm: Llama's Simplified Layer Norm

Standard LayerNorm

- $\text{LN}(\mathbf{x}) = \gamma \cdot \frac{\mathbf{x} - \mu}{\sigma + \epsilon} + \beta$
- Subtracts mean μ , divides by std σ
- Learnable scale γ and shift β
- Used in BERT, GPT-2, original Transformer

RMSNorm (Llama, Mistral, Gemma)

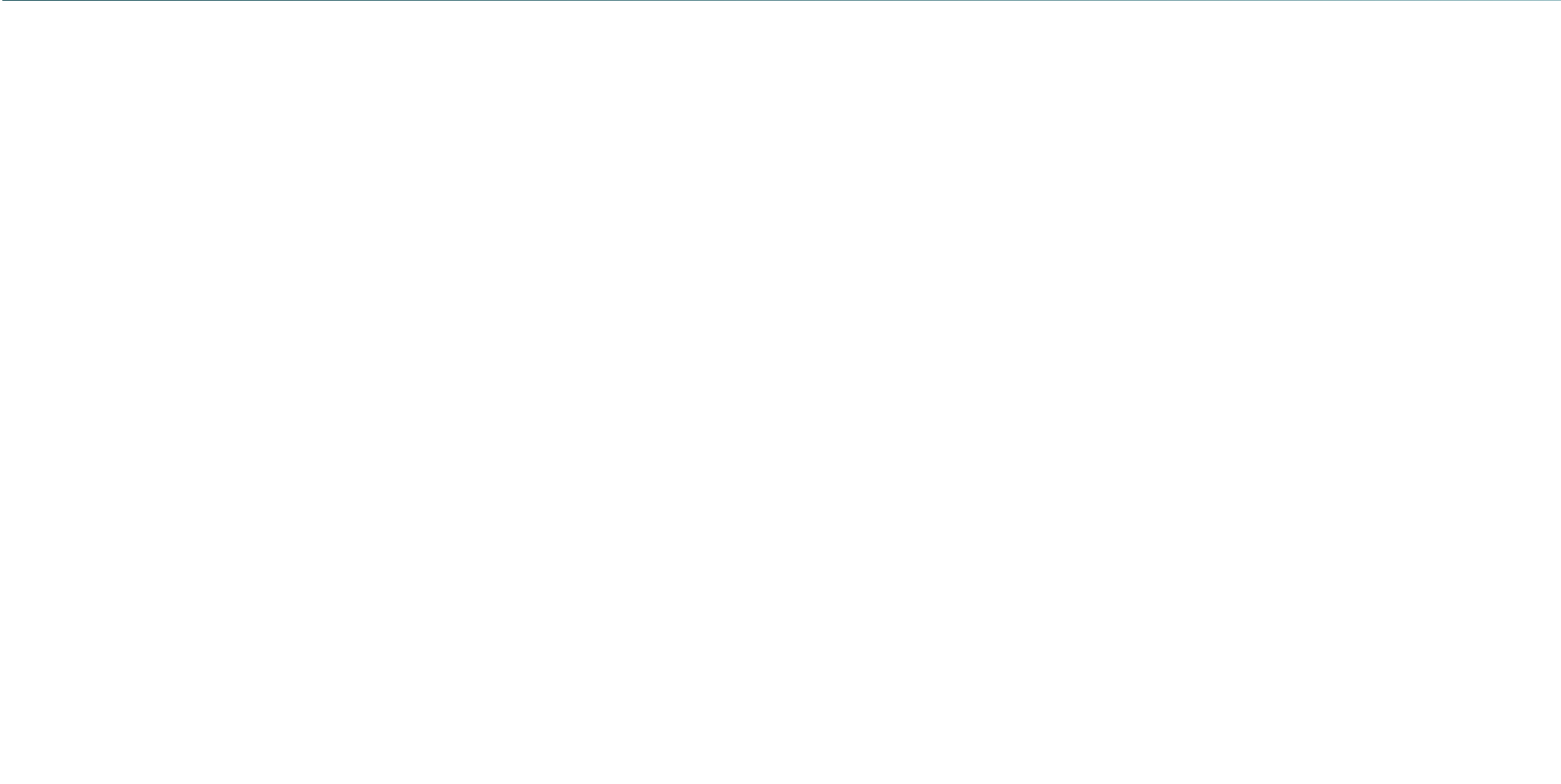
- $\text{RMS}(\mathbf{x}) = \gamma \cdot \frac{\mathbf{x}}{\sqrt{\frac{1}{d} \sum x_i^2 + \epsilon}}$
- No mean subtraction; only RMS-based scaling
- 7–15% faster than LayerNorm
- Comparable quality; slightly simpler gradient
- Proposed by Zhang and Sennrich (2019)



Full Llama Block: Residuals + RMSNorm

The Complete Llama Transformer Block (Pre-LN, RMSNorm)

- **Step 1:** $h = x + \text{Attention}(\text{RMSNorm}(x))$
- **Step 2:** $x' = h + \text{FFN}(\text{RMSNorm}(h))$
- **Attention:** Grouped Query Attention with RoPE positional encoding
- **FFN:** SwiGLU three-matrix projection (gate, up, down)
- **Final:** RMSNorm applied to the output of the last block before the LM head
- Residuals ensure that if any layer produces poor representations, earlier layers' features bypass it





GenAI Application: Training 96-Layer Models

How Residuals Enable Extreme Depth

- GPT-4 is estimated at 96–120 transformer layers; Gemini Ultra uses a Mixture-of-Experts with comparable depth
- Without residual connections, training gradients would vanish within 10–15 layers, making such models untrainable
- With Pre-LN residuals, gradients flow through all layers via the identity skip paths simultaneously
- **Gradient checkpointing** (recomputing activations in backward pass) is enabled by residuals: each block's input is stable enough to recompute from
- All large-scale distributed training frameworks (Megatron-LM, DeepSpeed) assume Pre-LN residual structure



GenAI Application: Probing Residual Stream

The Residual Stream Interpretation

- Mechanistic interpretability research frames the sequence of residuals as a **“residual stream”**
- Each layer *reads* from and *writes* to this stream: $\mathbf{x}_{l+1} = \mathbf{x}_l + \Delta_l$
- Different layers specialize: early layers handle syntax, middle layers coreference, late layers semantics and factual recall
- **Logit lens (Nostalgebraist, 2020)**: Apply the LM head at each intermediate residual to observe how predictions evolve layer by layer
- Used by Anthropic, EleutherAI, and DeepMind to understand and improve model internals

The residual stream interpretation is the foundation of circuit-level mechanistic interpretability, where researchers trace how specific information propagates through the stream across layers.



GenAI Application: Layer Pruning via Residuals

Shallow Decoder Distillation

- Because residuals create a stable stream, individual layers can be **removed** without catastrophic failure
- ShortGPT (2024): Drop the 50% least important layers from Llama 2 70B, retaining 80% of performance
- The remaining residual stream carries sufficient information

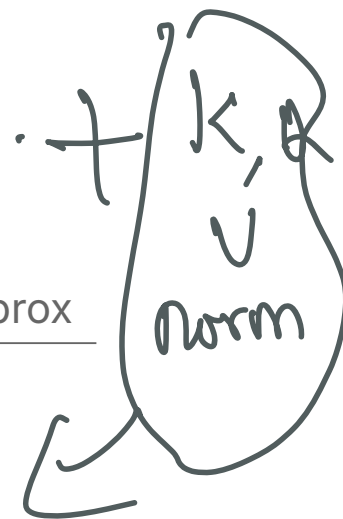
Practical Benefits

- Reduce inference latency by $2\times$ with $<5\%$ quality loss
- DistilBERT (Hugging Face): 6-layer BERT from 12 layers via layer-level residual distillation
- LaCo, SliceGPT: structured pruning targeting projection weights in residual blocks

Residual Placement Across Major Models



Model		Layers	Norm Type	Placement	Activation
Original	Trans-	6	LayerNorm	Post-LN	ReLU
former					
BERT-Large		24	LayerNorm	Post-LN	GELU
GPT-2 Large		36	LayerNorm	Pre-LN	GELU
Llama 2 / 3		32-80	RMSNorm	Pre-LN	SwiGLU
Mistral 7B		32	RMSNorm	Pre-LN	SwiGLU
Gemma 2 9B		42	RMSNorm	Pre-LN	GELU approx



Common Misconceptions



Myths and Corrections

- **Myth:** "Residual connections are only needed for very deep networks." Even shallow Transformers (6 layers) benefit: residuals improve convergence speed and final quality
- **Myth:** "Residuals simply copy the input and ignore the sub-layer." At initialisation, yes; but trained sub-layers learn meaningful corrections $F_l(\mathbf{x}_l)$
- **Myth:** "Pre-LN and Post-LN produce the same final model." They converge to different quality points; Pre-LN is consistently more stable and often better
- **Myth:** "RMSNorm is just a simplified LayerNorm with negligible effect." The 7–15% speed improvement is meaningful at billion-token training scale

Key Takeaways



What You Have Learned

- Residual connections add the input directly to the sub-layer output: $\mathbf{x}_{l+1} = \mathbf{x}_l + F(\mathbf{x}_l)$
- The identity term in the gradient $(1 + \partial F / \partial \mathbf{x})$ prevents vanishing gradients regardless of depth
- Pre-LN (norm inside the residual branch) is the modern standard; it provides training stability without warmup requirements
- Modern LLMs (Llama, Mistral, Gemma) use RMSNorm instead of LayerNorm for 7–15% training speedup
- The residual stream enables mechanistic interpretability, layer pruning, and incremental representation building
- Without residual connections, 96-layer GPT-4 scale models would be untrainable

Thank You

Residual Connections — Module 3.4